

legoESM: modular, differentiable, multiscale, AI-ready Earth system model

Pierre Gentine^{1,2,*}, Dhruv Balwada^{2,3}, Linnia Hawkins^{1,2}, Aytaç Paçal⁴, Alistair Adcroft⁵, Jianing Fang³, Aya Lahlou^{1,2}, Kara D. Lamb^{1,2}, David M. Lawrence⁶, Julien Le Sommer⁷, Joseph Mouallem⁸, Duncan Watson-Parris⁹, and Veronika Eyring⁴

¹Department of Earth and Environmental Engineering, Columbia University, New York, NY, USA

²Learning the Earth with Artificial Intelligence and Physics (LEAP) NSF Science and Technology Center, Columbia University, New York, NY, USA

³Lamont–Doherty Earth Observatory, Columbia University, Palisades, NY, USA

⁴University of Bremen, Institute of Environmental Physics, Bremen, Germany

⁵Program in Atmospheric and Oceanic Sciences, Princeton University, Princeton, NJ, USA

⁶National Center for Atmospheric Research, Boulder, CO, USA

⁷Institut des Géosciences de l’Environnement (IGE), Université Grenoble Alpes, CNRS, IRD, Grenoble INP, Grenoble, France

⁸NOAA Geophysical Fluid Dynamics Laboratory, Princeton, NJ, USA

⁹Scripps Institution of Oceanography and Halicioğlu Data Science Institute, University of California San Diego, La Jolla, CA, USA

*Corresponding author: pg2328@columbia.edu

Abstract

Earth system models (ESMs) have grown enormously in realism and process complexity, yet persistent biases and a stubbornly wide spread in the climate response to greenhouse-gas forcing remain, and the dominant modelling paradigm makes it hard to adopt three transformative tools: machine learning, gradient-based calibration, and accelerator hardware. Most ESMs are written in Fortran or C, so machine-learning components are implemented through one-way bridges that preclude online training; the absence of model adjoints forces derivative-free tuning; and only isolated components have been ported to GPUs. Further, the model hierarchy has driven much of our climate understanding, yet at present a separate model is used at each complexity level, without a single unifying infrastructure permitting clear hypothesis testing. Finally, models are built for either weather or climate time scales. We present legoESM, an Earth system model developed from scratch with AI agents and written entirely in JAX, in which every component (atmosphere, ocean, land, sea ice, coupler) is composable, with dynamical cores, physics schemes, grids and complexity levels swapping as interchangeable bricks, end-to-end differentiable, and hardware-portable (CPU, GPU, TPU). The same code base spans the hierarchy of models, across space and time scales, from large-eddy and cloud-resolving simulation through single-column and idealized cases to global atmosphere and ocean integrations, spanning weather to climate time scales. We demonstrate the flexibility of the model parameterizations against a cloud-resolving oracle, tuning classical physics or column neural networks coupled to a differentiable dynamical core with a spherical neural operator emulating the entire atmosphere, global gradient-based calibration of land-surface temperature and albedo bias, and grid-swapping validation against established models (NEMO, FV3) across cubed-sphere, lat–lon, icosahedral/Voronoi and tripolar grids. legoESM was developed almost entirely by AI coding agents working under a human-specified scientific contract, with mechanical verification of every physical invariant through systematic benchmarking. It is intended to provide an open, community-based playground for rapid prototyping, hypothesis testing, research and teaching.

1 Introduction

2 Earth system models have improved substantially over recent decades, both in their fidelity with
3 respect to the observed climate and in the breadth of processes they represent^{1,2}. Yet important
4 biases persist, so that the uncertainty in the response of the climate system to greenhouse-gas forcing
5 remains large and only loosely constrained by the historical record. Equilibrium climate sensitivity
6 is still uncertain at the $\sim 50\%$ level³; projections of rainfall extremes⁴, of ocean heat uptake, and of
7 the strength of the land carbon sink^{5,6} differ substantially across models and are often inconsistent
8 across different regions of the world. Further, climate models have to a large extent not embraced
9 the data revolution: only a fraction of the wealth of observations and observational products
10 (satellites, reanalyses) is used for model calibration and evaluation. This is often a reflection of the
11 difficulty of using modern AI methods, especially gradient-based ones, for model calibration.

12 Three developments could, in principle, reduce these uncertainties, but each is held back by
13 the prevailing software paradigm. First, machine learning (ML) has been used to emulate or
14 replace expensive parameterizations such as deep convection^{7–10} and ocean mesoscale eddies^{11,12},
15 often improving key processes such as rainfall extremes and eddy fluxes, and demonstrably reducing
16 systematic errors in cloud cover, precipitation and circulation^{13–15}. But because ESMs are typically
17 written in Fortran or C, ML components are coupled through one-way bridges that feed a neural-
18 network prediction into the physics model¹⁶. As a result, online training, that is learning through
19 the model’s own evolution, is generally impossible, and the resulting hybrid systems are prone
20 to instability⁹ and cannot be further tuned to observations. Second, many modelling centres
21 have recently focused on automated tuning and calibration of uncertain parameters^{14,17,18}, given
22 the growing complexity of the models, but because the models typically do not have an adjoint
23 (a gradient with respect to the parameters), calibration relies on derivative-free Monte-Carlo or
24 ensemble methods^{19–21} that scale poorly with the number of parameters. Third, only a handful of
25 models or components have been ported to GPUs^{22,23}, and this remains the exception rather than
26 the norm. This has limited the capability to go to higher resolution to better resolve key processes
27 such as deep convection²⁴.

28 A separate, long tradition in climate science is the concept of a hierarchy of models, spanning
29 idealized to complex configurations^{25–27}. This hierarchy has been fundamental to building physical
30 understanding of the climate system. Today, however, different classes of problems are addressed
31 with different, largely incompatible codes and models, so there is no unified framework in which to
32 test hypotheses systematically across scales with clean counterfactuals.

33 In parallel, a second revolution has transformed weather forecasting: data-driven models trained
34 on decades of reanalysis now rival or surpass physics-based forecasts at a tiny fraction of their
35 computational cost^{28–30}. Yet purely learned forecasts tend to drift and lose physical realism at
36 long lead times³¹, and on their own they do not provide the conserved, multi-component system
37 that climate projection demands. More fundamentally, machine-learning emulators, however skilful
38 and fast, reproduce input–output relationships without representing the mechanisms that gener-
39 ate them, which limits their value as scientific instruments. Earth system models are not merely
40 predictive tools but research tools: their capacity for hypothesis testing and mechanistic attribu-
41 tion is central to scientific progress. Preserving that capability while gaining the computational
42 and optimization advantages of differentiable, ML-enhanced methods is the central motivation for
43 process-based, differentiable Earth system models rather than pure emulation.

44 Finally, all current Earth system models are based on languages (Fortran, C, Julia) that are
45 not widely known by junior scientists. This inherently limits adoption, progress and rapid model
46 evolution. The recent weather AI revolution has demonstrated that it is not only model per-
47 formance that drives widespread adoption, but also ease of use. The confluence of large-scale

datasets, GPU-accelerated computing, and differentiable programming frameworks capable of scaling gradient-based optimization enabled the machine-learning revolution; Earth system science is now positioned to leverage the same convergence, but doing so requires Earth system models written in modern, automatically differentiable languages that can natively use GPUs, support gradient-based optimization, and interface directly with machine learning.

We argue that a modern ESM should be (i) easy to use, ideally written in Python, the lingua franca of junior scientists; (ii) hardware-agnostic, running on CPUs, GPUs and TPUs as accelerators evolve; (iii) composable, so that different numerical methods, physical parameterizations or entire modules (for example the whole atmosphere) can be swapped, combined and trained jointly for controlled experiments; (iv) able to integrate AI seamlessly, whether by replacing a single parameterization or an entire component (for example swapping the whole atmosphere for a neural network, as in recent AI weather models^{28–30,32}), which requires the code to be end-to-end differentiable so it can be trained on diverse data with modern ML optimizers; (v) solvable on many grids tailored to the problem at hand; and (vi) able to run across scales in large-eddy (LES), cloud-resolving (CRM), regional or global configurations, bridging spatial scales as well as temporal scales from weather to long-term climate.

A growing ecosystem of differentiable geoscientific models has begun to demonstrate these ideas, though so far one component or one scale at a time. For the atmosphere, differentiable dynamical cores and intermediate-complexity models such as the JAX Circulation Model³³, the neural general circulation model NeuralGCM³² and SpeedyWeather.jl³⁴ expose gradients for calibration, sensitivity analysis and online learning. For the ocean, JAX and automatic-differentiation efforts, from Veros³⁵ and jaxsw³⁶ to program-transformation approaches to differentiable ocean modelling³⁷, bring the same capability to circulation and state estimation. For the land surface and hydrology, differentiable models such as JAX-CanVeg³⁸ and differentiable process-based hydrology³⁹ train physical parameters and neural components end to end against observations, building on the broader case for differentiable programming in the geosciences^{40,41}. What no framework has yet provided is a single model that carries these properties across components, grids and scales at once, for the entire Earth system, rather than a differentiable treatment of one piece in isolation.

Here we present legoESM. Rather than a single Earth system model, it is an open, modular framework in which differentiable representations of individual processes can be assembled, coupled and benchmarked against alternative formulations to address diverse scientific questions, conceived from scratch as an evolving, agile, community-based infrastructure for testing ideas and hypotheses across weather and climate (Fig. 2). Distinctively, it was developed almost entirely with AI coding agents (Claude and Codex) operating under a human-authored scientific specification, with every declared physical invariant checked mechanically through contracts, faithfulness oracles and systematic benchmarking. We believe that legoESM can transform the way we do research and educate on weather and climate.

A model assembled from interchangeable bricks

legoESM was developed from scratch, in Python and JAX, using AI coding agents. It is built from interchangeable building blocks, or “bricks” (Fig. 1): each component can be developed, installed and run on its own, or combined with the others into a fully coupled model. Concretely, it is a *federation* of independently installable Python packages that share a single `legoesm` namespace and obey a strict one-way dependency graph. At the base sits the substrate, `legoesm-core`: physical constants, thermodynamics, the grid and operator library, state containers, time integrators, the device/precision runtime, and the parallel layer (halo exchange and reductions). Four Earth-system

components (`atmosphere`, `ocean`, `land` and `ice`) depend only on the core and never on one another, so each can be installed and run on its own (`pip install legoesm-ocean` yields a working ocean model). Above the components, an orchestration layer provides the tile-based `coupler` and drivers, the `ml` package (spherical neural operators, learned physics, differentiable training and variational data assimilation), and `tools` (external forcing, diagnostics with conservation gates, and visualization). The whole stack is a pure pytree, so JAX’s function transformations (`jit`, `grad`, `vmap`, `scan`) apply through the complete coupled model. Writing the model in JAX, rather than a lower-level language, secures two key properties at once: identical code runs natively on CPUs, GPUs and TPUs and will compile to whatever accelerators emerge next, sidestepping the multi-year effort of retrofitting a legacy model to new hardware; and the entire coupled model is differentiable, so gradients are obtained automatically without a hand-written adjoint.

The framework is built on four independently varying axes (Fig. 1). The **grid** (cubed-sphere C-D, lat-lon C-grid, Gaussian/spectral, Voronoi/MPAS, and tripolar) is chosen by one factory and shared by every component. The **model complexity** runs along physical ladders (shallow-water $\rightarrow$ hydrostatic $\rightarrow$ nonhydrostatic atmosphere; fixed sea-surface temperature (SST) $\rightarrow$ slab $\rightarrow$ multi-layer $\rightarrow$ full three-dimensional ocean; single-column model (SCM), large-eddy simulation (LES) and cloud-resolving model (CRM)), so that one framework spans scales from metre-scale turbulence to the global circulation, orthogonal to the numerical discretization. The **physics** is composable: each component sums the tendencies of any chosen subset of process parameterizations, with `combined` = $\sum_i \text{process}_i$ guaranteed additive to machine precision by tests. The **fidelity** ranges from research-idealized to operational, following the standard atmosphere-only, ocean-only and coupled model intercomparison protocols (AMIP, OMIP and CMIP), encoded as test tiers that each add a stricter conservation gate. Mass is conserved by construction; energy and momentum are tracked diagnostically with explicit budget closure. The model currently provides more than twenty dynamical cores across five discretization families and more than thirty physics parameterizations (Supplementary Information, Tables S1–S3), together with a 3D Boussinesq ocean, a multi-layer land model with carbon cycle and multi-category sea ice. The ocean and land span degrees of complexity from slab and multi-layer slabs to a full three-dimensional Boussinesq ocean and a full Richards-equation land model, with or without a carbon cycle. The framework is modular by design, so entire modules (for example the atmosphere) can be replaced by machine learning, such as a spherical Fourier neural operator (SFNO), providing a ladder from a fully physics-based configuration to fully emulated components. Full governing equations and numerics are given in Supplementary Information §1. The source is openly available at <https://github.com/gentine/legoESM>.

Development by AI agents under a scientific contract. legoESM was built through a human-agent workflow we summarise here because it is integral to the result. We first used an agentic coding environment to define the framework and a written specification (a project “constitution”) stating the vision, the conserved quantities, the sign and unit conventions, the valid sets of schemes, the reference literature, and the acceptance criteria for each component: the human dictates the *logic*; the agent fills the *body*; and every declared invariant is checked *mechanically* so that violations fail loudly. Agents then wrote small-scale unit tests covering most of the code, followed by a ladder of increasingly stringent scientific benchmarks: standard dynamical-core tests of rising complexity (shallow-water Williamson and Galewsky cases, the Jablonowski–Williamson baroclinic wave, Held–Suarez, and DCMIP-style cases); parameterizations exercised in radiative–convective equilibrium (RCE) coupled to ocean or land; and finally full atmosphere-only and ocean-only integrations compared against established models such as the NEMO ocean model and the GFDL FV3

atmospheric core. Boundary-layer turbulence was validated against canonical LES cases (GABLS1, the Ekman layer and Wangara); convection schemes were evaluated and then tuned against cloud-resolving models; and microphysics was tested in LES and CRM runs against radiative-convective-equilibrium intercomparison (RCEMIP) oracles. Weak and strong scaling were iterated on by the agents, with runs on the Derecho and Levante supercomputers, and performance on multi-node message-passing (MPI) and GPUs was improved iteratively; for the hardest problems, such as multi-node scaling, grid-imprint artifacts at cubed-sphere panel edges and AMIP faithfulness, we used a persistent “Ralph” iteration loop¹ to keep the agent working until the artifacts were eliminated. Throughout, an adversarial code-review agent (“codex”) reviewed every substantial change, which accelerated convergence and reduced defects; community agent-skill libraries were used in the final stages to reach operational quality and to surface subtle bugs. The result is a $\sim 10^5$ -line code base whose physical contracts are enforced by continuous-integration tripwires (Methods).

Two features of this workflow are worth emphasising, because they may matter more than legoESM itself. First, it is fast: with agents handling implementation and testing, adding a new parameterization or dynamical core, historically a months-to-a-year undertaking in an established model, can shrink to days, compressing the loop between a scientific idea and a tested, benchmarked implementation. Second, it is *expert-in-the-loop*: the agents are productive only because domain scientists specify the governing logic, conserved quantities and acceptance criteria and adjudicate the physical choices at each step. Without that guidance, a general-purpose coding agent would not, on its own, arrive at a coupled model of this scientific complexity with defensible numerics.

Many grids, one set of equations

A central design goal is that the governing equations be solvable on whichever grid best suits the problem. legoESM implements shallow-water, hydrostatic primitive-equation and nonhydrostatic compressible-Euler dynamics across the major grid families used in Earth system modelling: the cubed-sphere and latitude-longitude finite-volume grids, Gaussian/spectral transforms, and unstructured Voronoi (MPAS) meshes. The specific discretizations, transport operators and their community lineages are listed in Supplementary Table S1. As an illustration, the same shallow-water equations solved on the cubed-sphere, lat-lon, icosahedral and spectral grids give near-identical solutions across all four discretizations, including for the propagating Rossby-Haurwitz wave (Williamson 6)⁴², and the cubed-sphere C-D core reproduces the Jablonowski-Williamson baroclinic wave⁴³. Removing a residual grid imprint at cube-panel boundaries, visible only on the meridional wind and not in error norms, was one of the problems for which the agentic “Ralph” iteration loop proved essential; a duo-grid treatment⁴⁴ eliminates it. The full cross-grid verification suite (cosine-bell advection, the Williamson set, the baroclinic wave and convergence rates) is given in Supplementary Information §2.

Composability and differentiable calibration

Swapping one physical scheme for another to assess its effect is a routine scientific need, yet it is difficult in conventional climate models, in which parameterizations are interwoven with the rest of the code. In legoESM, every parameterization presents the same tensor-in/tendency-out interface, taking the atmospheric state and returning a tendency, so schemes can be exchanged without touching the rest of the model. Figure 3 illustrates this for atmospheric convection, the moist rising

¹The “Ralph” technique runs an agent repeatedly on the same task, feeding test and diagnostic results back each cycle, until a target criterion is met (<https://ghuntley.com/ralph/>).

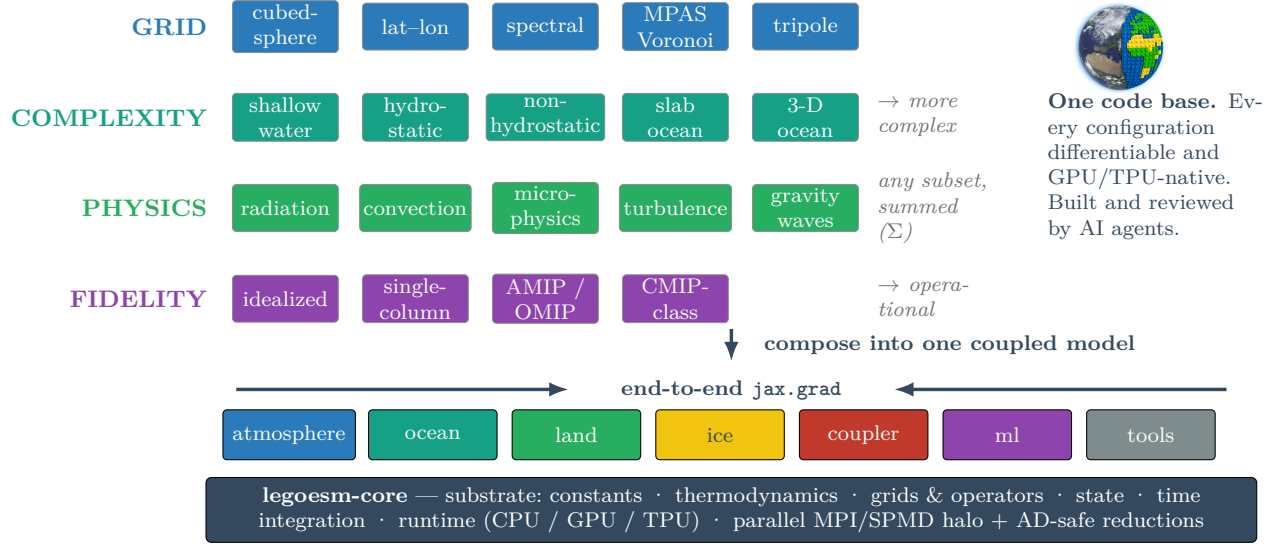


Fig. 1: **One code base, composable along four independent axes and differentiable end-to-end.** Top: any numerical *grid*, level of *complexity*, set of *physics* schemes and evaluation *fidelity* is selected independently and combined by changing a few lines of configuration; complexity and fidelity run along ladders of increasing realism, while physics tendencies are additive to machine precision (Σ), so processes switch on and off freely. Bottom: the selected bricks assemble into one coupled model, a federation of independently installable packages that share the **legoesm-core** substrate and obey a strict one-way dependency graph (`pip install legoesm-ocean` yields a working ocean model). Because the whole stack is a pure JAX program, exact gradients propagate through the entire coupled model (`jax.grad`) and identical code runs on CPU, GPU and TPU. Package contents and the full scheme inventory are given in Methods and Supplementary Information (Tables S1–S3).

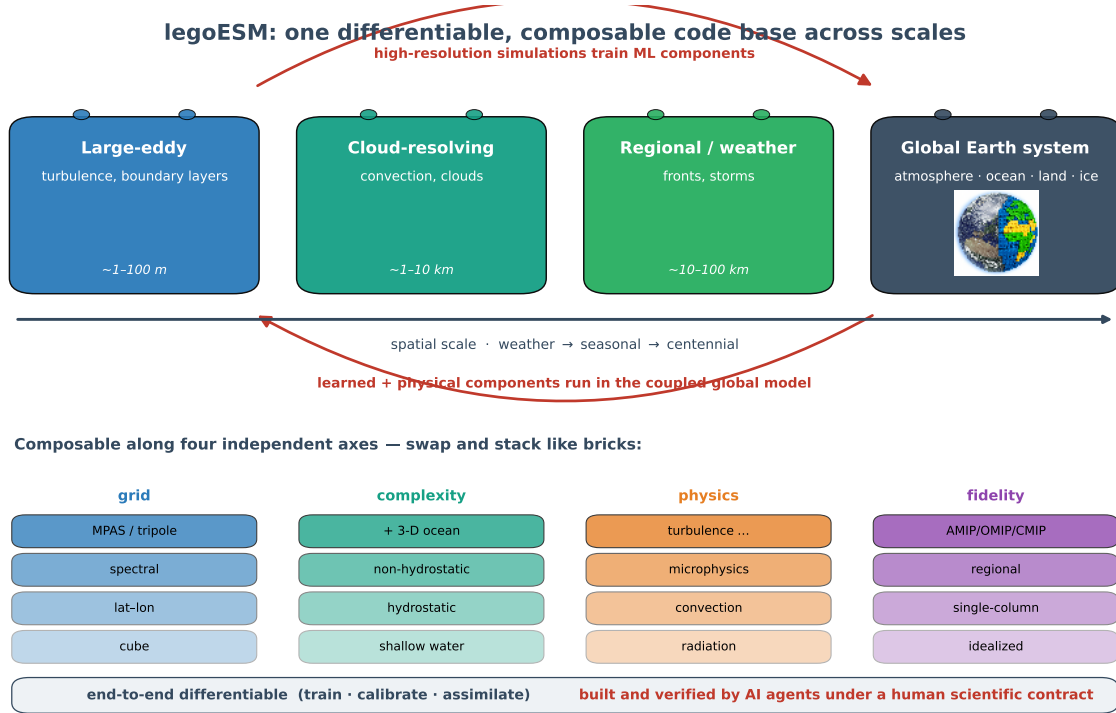


Fig. 2: One code base across scales, composable along four axes. The same legoESM bricks run from metre-scale large-eddy simulation, through cloud-resolving and regional configurations, to the global coupled Earth system (top), and span weather to centennial timescales. High-resolution simulations supply training targets for machine-learned components, which are then deployed within the coupled global model (red loop). Any configuration is assembled by choosing independently along four axes (bottom): the computational grid, the physical complexity, the set of physics schemes, and the fidelity tier. The whole stack is differentiable end to end and was built and verified by AI agents under a human-defined scientific specification.

motions that produce most tropical rainfall and a long-standing source of model error. Ten different convection schemes, spanning the mass-flux, adjustment and stochastic families (Extended Data Table 2), are each run, unchanged, in a single atmospheric column held in radiative-convective equilibrium (a small, well-understood testbed) and scored against the temperature profile that theory predicts. They separate cleanly by skill (Fig. 3), recovering the expected ordering and exposing which ones fail to reach equilibrium: an experiment that is cumbersome in a conventional model but a one-line change here. The same mechanism can replace an entire component: a classical physics suite, a column neural network, or a full spherical neural operator⁴⁵ (a learned dynamical core) can occupy the same slot, and the ocean can run as a prescribed sea surface, a simple slab, or a full three-dimensional model, all from one driver (Supplementary Information §3).

Differentiability turns calibration into gradient-descent optimization, leveraging one of the key backbones of the machine-learning revolution. Because gradients flow through the model via `jax.grad`, parameters can be optimized directly against data with modern ML optimizers (Adam, MUON), rather than by derivative-free search^{17,19} that scales poorly in high dimension. Figure 4 demonstrates this for the land surface: tuning per-plant-functional-type surface parameters by gradient descent against ERA5⁴⁶ mean-annual skin temperature, under physical-bound constraints imposed through differentiable transforms, reduces the global mean warm bias from +1.09 to +0.30 K, with the largest improvements over the previously too-warm arid and semi-arid land. The same machinery has been used to calibrate convection-scheme parameters against cloud-resolving references, and to adjust a cloud-fraction critical-relative-humidity threshold using gradients of the top-of-atmosphere shortwave and longwave fluxes so that the planetary albedo falls in its observed ~30% range. The differentiable parallel layer makes this work at scale: exact reverse-mode gradients flow through distributed halo exchange (via a custom adjoint that swaps send/receive ranks) and through the global reductions used by the mass-conservation fixers (Methods).

The same adjoint capability underpins variational data assimilation. A three-dimensional variational (3D-Var) analysis is implemented for the atmosphere: the analysis increment is obtained by differentiating the observation operator and the background term through the model, and it extends to 4D-Var through the differentiable time-stepping kernel, with no separately coded adjoint model. Because calibration, data assimilation and hybrid training all reduce to taking gradients through the *same* model, legoESM makes it natural to pose questions that are cumbersome in conventional models, for example which processes a given bias is most sensitive to. As a concrete example of the capability we most want to highlight, the gradient of any scalar diagnostic J (a global-mean radiative imbalance, a regional precipitation bias, a measure of tropical variability) with respect to *every* internal parameter and boundary input is obtained in a single reverse pass, at a cost essentially independent of the number of parameters; ranking those sensitivities attributes the diagnostic to the processes that control it, where the brute-force equivalent would demand thousands of perturbed integrations. This shifts the emphasis from demonstrating predictive skill to using the model as an instrument for understanding.

Bridging scales and the model hierarchy, from turbulence to the globe

Climate processes span an enormous range of scales, from turbulent eddies metres across to circulations spanning the planet, and traditionally each scale is treated with a separate model. legoESM covers this range within a single framework. On a small doubly periodic domain, its nonhydrostatic core with a turbulence closure⁴⁷ performs large-eddy simulation, resolving the largest turbulent motions directly: the canonical GABLS1 stable boundary layer⁴⁸, the Ekman layer, the Wangara

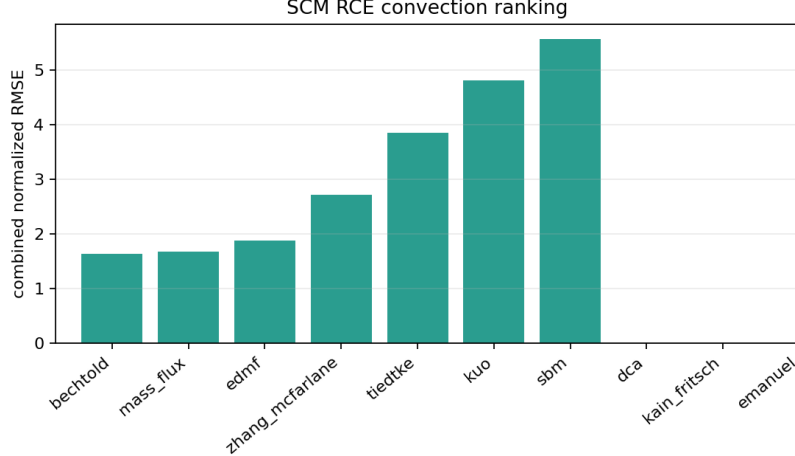


Fig. 3: **Composability: convection schemes swapped unchanged.** Each of ten convection parameterizations is run, unchanged, in a single atmospheric column in radiative–convective equilibrium and scored by how closely its equilibrated temperature profile tracks the surface moist adiabat that theory predicts. Ranking the schemes by this combined normalized error separates them cleanly by skill, with the mass-flux, EDMF and Bechtold family clustering at low error; schemes that do not reach a realistic equilibrium in this idealized single-column setup are left without a score. Selecting a scheme is a one-line configuration change. The full per-scheme temperature-profile sweep is shown in Supplementary Information §3, and the complete scheme inventory is in Extended Data Table 2.

diurnal cycle, and the DYCOMS-II⁴⁹ and RICO⁵⁰ cloud-topped layers all reproduce the expected mean profiles and turbulence structure (Supplementary Information §4). Enlarging the domain and adding moist physics turns the same core into a cloud-resolving model: in radiative–convective equilibrium⁵¹ it spontaneously organizes convection into clusters (“self-aggregation”), and its warm-rain microphysics matches a benchmark particle-based reference^{52,53} (Supplementary Information §4). Turbulence and microphysics schemes were additionally checked against well-regarded LES codes (MicroHH) and the cloud-resolving configuration against SAM. At the global scale, the cubed-sphere atmosphere runs prescribed-ocean (AMIP-style) simulations, evaluated with ESMValTool¹, and the three-dimensional ocean runs forced (OMIP-style) integrations (next section). Spanning a metre-scale eddy to a global simulation within one differentiable code base is, to our knowledge, unique, and it is precisely what hybrid physics–AI modelling needs, since the high-resolution end can generate training data and the global end can host the learned scheme.

Global simulations against established models

A stringent test of the framework is whether its ocean component, built from the same composable elements, can reproduce a mature community model on the complex grids used in operational ocean modelling. The three-dimensional Boussinesq ocean uses a split-explicit baroclinic/barotropic solver⁵⁴ with KPP vertical mixing⁵⁵ and GM–Redi eddy parameterization⁵⁶, and runs on latitude–longitude, cubed-sphere, unstructured (MPAS/Voronoi) and NEMO-style tripolar grids. Figure 5 compares legoESM ocean integrations against NEMO⁵⁷ on two of the most demanding meshes. On the unstructured MPAS/Voronoi icosahedral grid, the simulated sea-surface temperature and

Mean-annual land surface: ERA5 reference and model biases (CLM default vs ERA5-tuned)

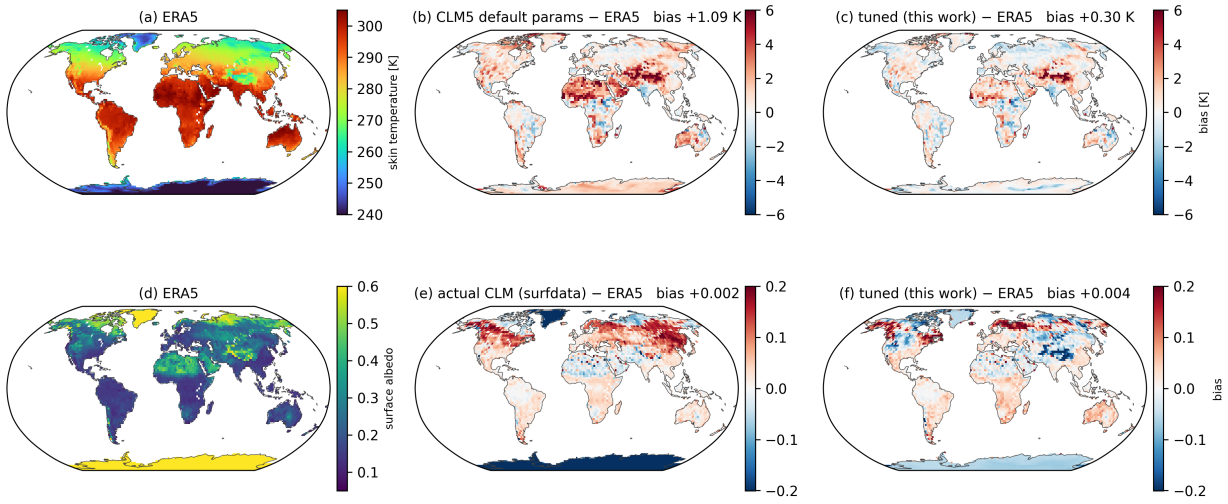


Fig. 4: **Differentiability: gradient-based calibration of the land surface.** Per-plant-functional-type surface parameters are tuned by gradient descent *through the differentiable land model* against ERA5⁴⁶ mean-annual fields, with physical bounds imposed by differentiable (sigmoid) transforms. Top: skin temperature. Starting from CLM5 default parameters (b), the global mean warm bias against ERA5 (a) falls from +1.09 to +0.30 K after tuning (c), with the largest gains over arid and semi-arid land. Bottom: surface albedo (d–f) stays essentially unbiased through the calibration (+0.002 to +0.004), confirming that the temperature fit is not bought at the cost of the radiative surface state. Gradients are obtained by reverse-mode differentiation of the land model; experimental set-up is in Supplementary Information §5.

the zonal-mean temperature section closely track NEMO across the global domain and the full water column (Fig. 5a,b). On the tripolar grid, whose folded northern boundary is a classic source of numerical artifacts, the modelled sea-surface height matches the NEMO field with a spatial correlation of 0.95 and near-zero bias (Fig. 5c). A centennial spin-up library, an AMOC tracker, conservation diagnostics, and a peer-comparison fidelity harness support quantitative model intercomparison (Supplementary Information §5). That an agent-built ocean reproduces a mature community model on its own native, geometrically awkward grids is a strong test of the framework. The same drivers run the standard uncoupled model-intercomparison protocols end to end. In atmosphere-only (AMIP) mode, the cubed-sphere atmosphere is integrated with prescribed observed sea-surface temperatures and sea ice and reproduces a realistic global climatology of near-surface temperature, precipitation and large-scale circulation. In ocean-only (OMIP) mode, the three-dimensional ocean is driven by prescribed surface fluxes and tracks NEMO, as shown above. In land-only (LMIP) mode, the land model is forced by reanalysis meteorology; this is the configuration used for the differentiable calibration of Fig. 4. All three are described, with their forcing and evaluation metrics, in Supplementary Information §5.

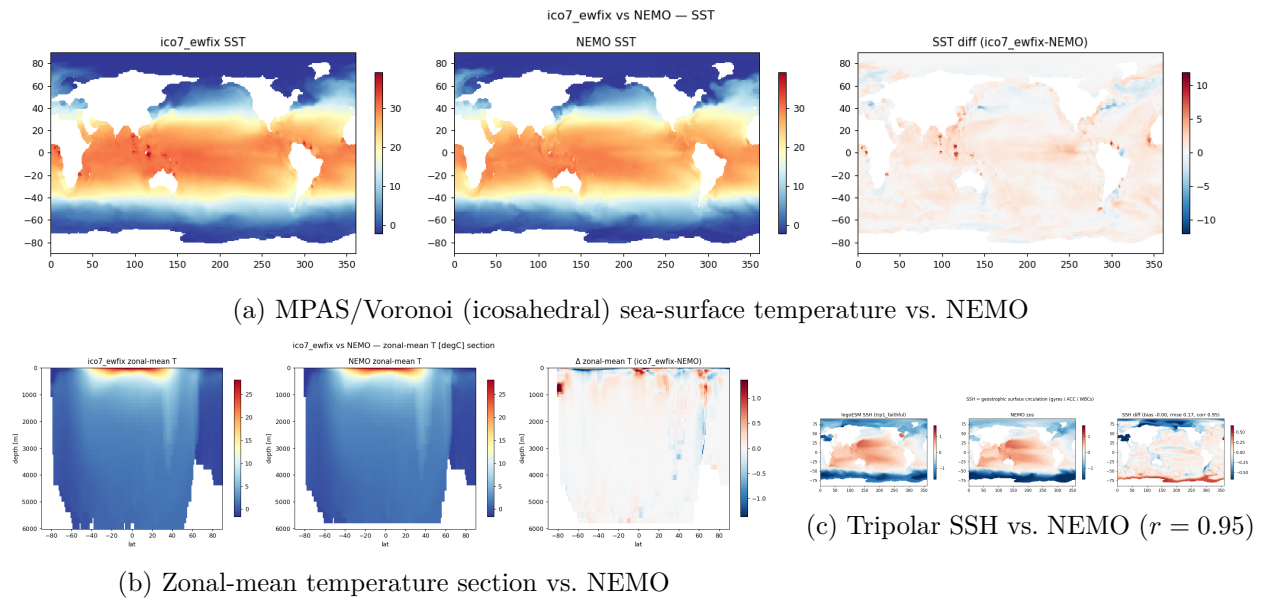


Fig. 5: A global ocean validated against NEMO on complex grids. (a) Sea-surface temperature and (b) zonal-mean temperature section from the legoESM ocean on the unstructured MPAS/Voronoi icosahedral grid, against NEMO, with difference maps. (c) Sea-surface height on the NEMO-style tripolar (eORCA) grid reproduces the NEMO field with spatial correlation $r = 0.95$ and bias ≈ 0 , despite the folded northern boundary. Full ocean diagnostics, additional grids and the fidelity harness are in Supplementary Information §5.

Performance and scalability

Long integrations and large ensembles demand efficiency on whatever hardware is at hand, and writing legoESM in JAX delivers this portably, without hand-written accelerator code. A single consumer GPU accelerates the atmosphere by 16–21 $\times$ and the ocean by 33–99 $\times$ over a 24-core CPU at single precision (Supplementary Information §6), with structured grids (lat–lon, cubed-sphere) outperforming unstructured (MPAS) and spectral cores, as expected from memory-access patterns.

Two algorithmic changes found during agentic optimization dominate the gains: parallelizing the correlated- k radiation over its g -points ($\sim 6\times$ on the radiation kernel) and a split-explicit barotropic ocean solver that escapes the global-reduction latency wall of the implicit solver (up to $1.65\times$ faster at 64 ranks). Across MPI ranks and devices the model scales well where communication is amortized (Fig. 6): the ocean reaches near-ideal parallel efficiency at production tile sizes, and the icosahedral atmosphere retains 0.69 parallel efficiency at 32 ranks (relative to an 8-rank baseline). Reverse-mode differentiability is preserved across all of this: gradients flow correctly through MPI halo exchange and global sum reductions, verified against single-rank references to machine precision (Methods). Honest limits are documented (spectral transforms remain communication-bound, and cross-node cube-panel exchange on a commodity TCP fabric anti-scales) and are reported in Supplementary Information §6 rather than hidden.

Accessibility, physical consistency and reproducibility

Because legoESM is written in idiomatic Python, an entire experiment is a few lines of readable code (selecting a grid, a complexity rung, a set of physics schemes and a forcing), which lowers the barrier for students and domain scientists (Supplementary Information §7). Physical consistency is treated as a first-class, mechanically tested property rather than an aspiration. Operators reproduce their analytic targets: the vector-invariant advection (Burgers), Laplacian viscous decay, Coriolis and inertial-oscillation tendencies all match closed-form solutions, and finite-volume transport and gravity-wave terms converge at their design orders under CFL-controlled refinement (Supplementary Information §8). Every physics module declares a machine-readable contract of its units, signs, conserved quantities and differentiability, and continuous-integration tripwires reject any change that hard-codes a physical constant outside the central registry, re-derives a saturation curve, silently drops a scheme on a typo, or breaks additive composition (Methods). Every run emits a reproducibility manifest (resolved configuration, state digest, git commit and library versions) so that a single command re-executes it bit-for-bit.

Summary and way ahead

legoESM shows that a single, high-level, differentiable code base can span the Earth-system model hierarchy, from large-eddy and cloud-resolving simulation to global atmosphere and ocean integrations, while remaining composable enough to swap dynamical cores, grids, physics schemes and complexity levels as interchangeable bricks, and differentiable enough to calibrate parameters and train embedded neural networks by gradient descent through the model. The single property that makes learning inside the model possible, differentiability from end to end, also delivers gradient-based calibration and exact data assimilation, drawing three previously separate research programmes into one framework. Concretely, the architecture makes practical several things that are awkward or impossible in a conventional model: differentiating a whole-model diagnostic back to the parameter or input that controls it (end-to-end sensitivity and attribution); training an embedded neural component online, through the model’s own evolution rather than on offline single-step tendencies; and running the identical code from a large-eddy column to a global simulation for clean cross-scale comparison. These capabilities, rather than any single simulation, are the novelty we wish to demonstrate.

Two broader points deserve emphasis. First, treating differentiability and composability as architectural invariants, rather than features bolted onto an existing model, is what makes the hierarchy continuous: the LES closure, the global parameterization and the neural operator are the

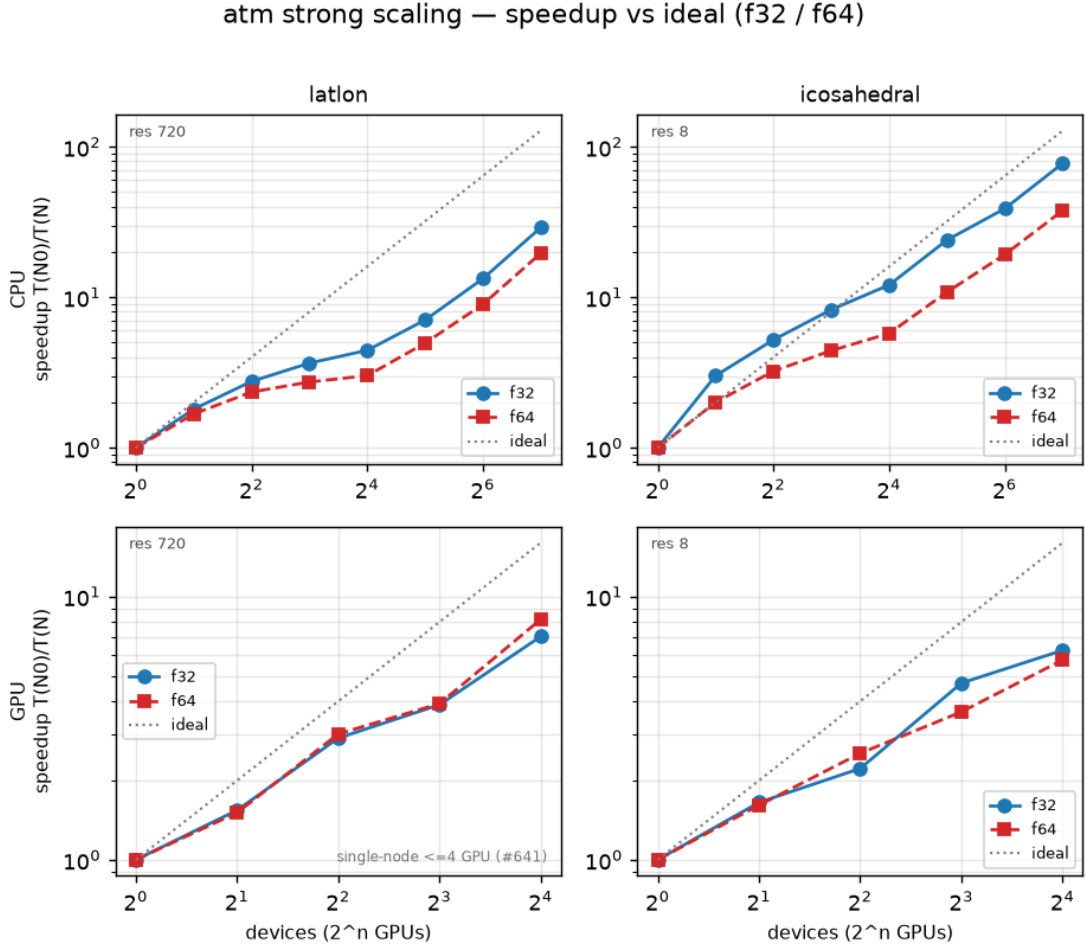


Fig. 6: **Strong scaling of the atmospheric dynamical core.** Speed-up relative to a single device, $T(N_0)/T(N)$, as processors are added: on CPU up to 128 MPI ranks (top) and on GPU up to 16 devices (bottom), for the lat–lon (left) and icosahedral (right) grids at single (f32) and double (f64) precision; the dotted line is ideal linear scaling. Scaling follows the ideal line while per-device work amortizes the halo exchange and rolls off once communication dominates, earlier on the GPU because each device already holds little work relative to the exchange. Absolute throughput and simulated-years-per-day for all grids and both hardware classes are in Supplementary Information §6.

same kind of object and obey the same interface. Second, the model was built largely by AI agents, but its trustworthiness rests on a human-authored scientific contract whose every clause is checked mechanically; the agent supplies implementation effort, while conservation, sign conventions, valid scheme sets and acceptance criteria are dictated and verified by people. We found adversarial agent review and persistent iteration loops essential precisely for the problems where physical intuition is hard to encode, such as grid-imprint artifacts at cubed-sphere edges.

A third point concerns the role of benchmarks. Agentic development of legoESM was possible only because the community has, over decades, distilled its understanding into open, well-posed benchmark cases and datasets, from the standard dynamical-core tests to RCEMIP, GABLS and the AMIP, OMIP and LMIP protocols, which gave the agents unambiguous targets against which every component could be checked. This hints at a broader shift: where a model’s source code was once the primary repository of physical understanding, in an agentic workflow the code becomes comparatively inexpensive to regenerate, and it is the tests, benchmarks and reference datasets, the accumulated work of the whole community, that increasingly encode what we know and that must be curated and preserved most carefully.

legoESM builds on the extraordinary body of physical understanding encoded in community models developed over decades (from the parameterizations of the Community Earth System Model⁵⁸, the ocean physics of NEMO⁵⁷, the dynamical cores pioneered at GFDL, and many others; see Extended Data Tables 1–10), repackaged in a modern, differentiable infrastructure. This paper is a demonstration of the potential of that architecture rather than a finished climate model: the framework has not yet been evaluated in long, fully coupled climate integrations, and so does not yet match the maturity of established operational models. Two caveats deserve emphasis. Gradient-based calibration fits a chosen target efficiently, but it does not remove structural error: a well-tuned configuration can still carry the biases inherent in its scheme set, and sharpening one metric can worsen others. And while the differentiable coupler makes assembling components straightforward, we anticipate the same hard coupled-system problems that challenge established models, such as spurious deep-water formation in the Labrador Sea, drift in the Atlantic meridional overturning circulation, and biases in tropical interannual variability (ENSO); demonstrating skill on these is essential future work. To make the framework usable and checkable by others, it ships a set of curated reference configurations, the idealized dynamical-core tests, radiative–convective equilibrium, and the AMIP, OMIP and LMIP set-ups shown here, each with expected results, so that a student or a new group can reproduce a known-good baseline before modifying it. Framed this way, legoESM is intended as an open, evolving research and educational tool, for theoretical insight, hypothesis testing and clean counterfactual experiments, and, in time and with sustained community contributions, potentially a full-fledged climate model. Such differentiable, AI-ready infrastructure is a concrete step toward the hybrid Earth system models that are increasingly seen as a route to sharper, more actionable climate projections²⁴. By making the full coupled system easy to read, easy to differentiate and easy to run on modern hardware, we hope to lower the barrier to experimentation and to invite the community to build on and contribute to it.

Finally, the strategy reaches beyond climate. The design choices underlying legoESM (composability, differentiability, hardware portability, and agent-assisted development under a human-defined scientific contract) are responses to structural features shared by many multiscale physical systems: combustion and reactive flows, materials science, plasma physics, astrophysical fluid dynamics, and biological transport all couple processes across vastly different scales, all stand to benefit from gradient-based inverse methods, and all face the same tension between legacy code and modern accelerators. A composable, differentiable, agent-built framework in which domain scientists specify the governing logic, conserved quantities and acceptance criteria while AI agents write and mechanically verify the implementation offers a template for simulation-based science far

more broadly, wherever physical models are complex, data are abundant and trustworthiness must be earned rather than assumed.

Methods

The complete governing equations and numerical methods of all components are given in Supplementary Information §1; this section summarises the computational implementation. End-to-end differentiability, conservation and stable-shape (non-retracing) JAX execution are treated throughout as hard constraints.

Agentic development methodology

legoESM was implemented by large-language-model coding agents in an agentic coding environment under a human-authored specification. Most of the code was written by Claude (Anthropic; the Opus and Sonnet model families) driving edits, test execution and debugging, and a second agent based on Codex performed adversarial review of every substantial change. A human-maintained specification fixed the conserved quantities, sign and unit conventions, valid scheme sets, reference literature and per-component acceptance criteria, and domain scientists reviewed and adjudicated the scientific choices at each step (expert-in-the-loop). For hard, slowly-converging problems (multi-node scaling; cube-edge grid artifacts) a persistent “Ralph” loop re-ran the agent on the same task, feeding back test and diagnostic output until the acceptance criterion was met; community agent-skill libraries (**agent-skills**, **superpowers**) were used in the final hardening stages. Physical contracts are enforced mechanically by continuous-integration tripwires: constants come only from a central registry, saturation curves only from the shared thermodynamics module, every scheme factory raises on an unknown selection, and additive physics composition is asserted to machine precision, each tripwire shipping a self-test that proves it is non-vacuous.

Use-case configurations. The convection sweep (Fig. 3) runs each scheme, unchanged, in the 40-level single-column model to a statistically steady radiative-convective equilibrium and scores the mean temperature profile against the surface moist adiabat. The land calibration (Fig. 4) optimizes per-plant-functional-type soil-water and surface parameters by gradient descent (Adam⁵⁹) against ERA5⁴⁶ annual-mean skin temperature, with physical bounds imposed by sigmoid transforms and gradients taken through the differentiable land integration. Ocean comparisons (Fig. 5) integrate the three-dimensional ocean on MPAS/Voronoi and tripolar grids against NEMO⁵⁷.

JAX implementation and differentiability. All components are pure-functional JAX, integrated in time with `lax.scan` and using `jnp.where/lax.cond` for data-dependent branching so that the program is differentiable and JIT-compilable. State objects and fields are registered as pytrees, so `jit`, `grad`, `vmap` and `scan` apply through the whole coupled model. Feature gates that must not trace both branches (e.g. conservation fixers) use a static Python conditional in a closure. Because buffer donation conflicts with reverse-mode autodiff, the compiled time-stepping kernel provides a non-donating variant for use inside `eqx.filter_value_and_grad`, and per-iteration forcings (insolation, SST) are passed as explicit traced arguments rather than captured in closures to avoid recompilation.

Differentiable distributed execution. Distributed runs use `initialize_distributed` to build the MPI topology, then scatter to rank-local tiles, step, and gather only for I/O. Halo exchange uses a four-dimensional path that sends all vertical levels in one message per neighbour. Reverse-mode gradients flow through communication: every `sendrecv` is wrapped in a custom vector-Jacobian product that swaps source and destination ranks in the backward pass, and the mass-conservation fixers differentiate through `allreduce(SUM)`. Only summation reductions are AD-safe; `max/min/allgather` are restricted to diagnostics. Multi-rank results are verified against single-rank references to $\sim 10^{-9}$, and gradients are checked the same way.

Training and calibration. Three training modes share one differentiable segment kernel: physics-parameter tuning, neural GCM, and SFNO-plus-dynamical-core. Trainable parameters are injected through the configuration pytree inside the loss (so leaves are traced), while production keeps static Python-float leaves that constant-fold without retracing. Physical bounds are imposed by sigmoid/softplus transforms. Optimization uses warmup-plus-cosine schedules with gradient clipping (Adam⁵⁹ or Muon⁶⁰); ERA5⁴⁶ is ingested through a Zarr-cached loader. The land calibration in Fig. 4 optimizes per-plant-functional-type parameters against ERA5 skin temperature; the convection calibration optimizes scheme parameters against a cloud-resolving reference.

Verification harness. The human-authored specification is enforced by mechanical tripwires in continuous integration: physical constants may come only from a central registry; saturation curves may come only from the shared thermodynamics module; every scheme factory must raise on an unknown selection rather than silently default; and additive physics composition is asserted to machine precision. Each physics module declares a contract (units, signs, conserved quantities, differentiability, reference, idealized test), and a self-test proves each tripwire is non-vacuous. Dynamical cores are gated on the standard sphere test set^{42,43,61}; spatial/grid artifacts additionally require visual regression because error norms are necessary but not sufficient. Substantial changes additionally passed an adversarial code-review agent.

Configurations for the figures. Shallow-water tests use cubed-sphere C36, lat–lon 72×144 , icosahedral level-5 and spectral T21 grids (Supplementary Information §2). RCE convection sweeps (Fig. 3) use the single-column model with shared physics. LES and CRM cases use the nonhydrostatic plane core, the latter following the RCEMIP RCE300 protocol⁵¹ (Supplementary Information §4). Ocean comparisons (Fig. 5) use MPAS/Voronoi (icosahedral) and tripolar (eORCA) grids against NEMO⁵⁷. Scaling was measured on a single RTX-class GPU (throughput/speed-up) and on a CPU cluster with one process per node (strong/weak scaling; Supplementary Information §6). Full settings are in Supplementary Information §1–§8.

Data availability

The ERA5 reanalysis is available from the Copernicus Climate Data Store. NEMO reference output and the idealized-test, LES, CRM and OMIP datasets used for the figures are available from the authors and via the model repository.

Code availability

legoESM is open source (MIT licence) and under active development at <https://github.com/gentine/legoESM>. A separate, version-frozen companion repository (legoESM v1.0), archived on Zenodo with a citable DOI on acceptance, contains the notebooks, configurations and reference data needed to reproduce every figure and experiment in this paper, so that the results remain reproducible independently of the evolving model code.

Acknowledgements

We thank Owen Hughes and Christiane Jablonowski for valuable discussions. legoESM builds on parameterizations and numerical methods developed over decades by the broader climate modelling community, including contributions from the CESM, GFDL, ECMWF, NCAR, CICE and NEMO development teams, and we gratefully acknowledge that heritage (Extended Data Tables 1–10). We also thank the CMIP community for the forcing datasets used in the atmosphere-, ocean- and land-only simulations. This work was supported in part by the U.S. National Science Foundation Science and Technology Center “Learning the Earth with Artificial Intelligence

and Physics” (LEAP). V.E. and A.P. were additionally supported by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) through the Gottfried Wilhelm Leibniz Prize (reference no. EY 22/2-1). legoESM was developed with the assistance of AI coding agents (Claude and Codex) operating under a human-authored scientific specification, and made use of the open-source `agent-skills` (<https://github.com/addyosmani/agent-skills>) and `superpowers` (<https://github.com/obra/superpowers>) libraries. This work used resources of the Deutsches Klimarechenzentrum (DKRZ), granted by its Scientific Steering Committee (WLA) under project ID no. 1179.

Author contributions

P.G. conceived the scientific vision, led the project, specified the scientific contract and wrote the paper. V.E. co-shaped the scientific vision and helped guide the modelling and evaluation approach. D.B., L.H. and A.P. led the OMIP, LMIP and AMIP developments and simulations, respectively. P.G., D.B., L.H., A.P., A.A., J.F., A.L., K.D.L., D.M.L., J.L.S., J.M., D.W.-P. and V.E. contributed to model design, component development, validation and analysis across the atmosphere, ocean, land, sea ice, coupler, machine-learning and high-performance-computing subsystems. All authors discussed the results and edited the manuscript.

Competing interests

The authors declare no competing interests.

Extended Data

legoESM re-implements, within a single differentiable framework, numerical methods and parameterizations developed by the weather, climate, ocean and machine-learning communities over many decades. Extended Data Tables 1–10 credit the principal schemes available in each component together with their community lineage; the lists are representative rather than exhaustive.

Extended Data Table 1 | Radiation schemes and community lineage.

Scheme	Description	Community lineage
Gray (Frierson)	Two-stream gray radiation with tunable optical depth	Frierson et al. (2006, JAS)
RRTMGP correlated- k	128 LW + 112 SW g -points; full gas optics	Pincus et al. (2019, JAMES); Mlawer et al. (1997, JGR); jax-rrtmgp (Parker, Watson-Parris, Nathaniel)

Extended Data Table 2 | Convection schemes and community lineage.

Scheme	Description	Community lineage
Simplified Betts–Miller	Relaxation to reference profile with CAPE trigger	Betts & Miller (1986, QJRMS); Frierson (2007, JAS)
Bechtold	Mass-flux deep/shallow with CAPE closure	Bechtold et al. (2001, QJRMS); ECMWF IFS lineage
Tiedtke	Mass-flux with moisture-convergence closure	Tiedtke (1989, MWR); ECHAM lineage
Kain–Fritsch	Mass-flux with dilute CAPE closure	Kain & Fritsch (1990/1993, JAS); WRF/NCAR
Zhang–McFarlane	CAPE-based mass-flux with downdrafts	Zhang & McFarlane (1995, Atmos.–Ocean); CESM/CAM
Emanuel	Spectral mass-flux with sub-cloud quasi-equilibrium	Emanuel (1991, JAS); MIT lineage
Kuo	Moisture-convergence closure	Kuo (1965/1974, JAS)
Deep convective adjustment	Column instability removal by T/q mixing	Manabe et al. (1965, MWR)
EDMF	Eddy-diffusivity/mass-flux unified BL–shallow	Siebesma et al. (2007, JAS); ECMWF/Siebesma lineage

Extended Data Table 3 | Microphysics schemes and community lineage.

Scheme	Description	Community lineage
Kessler	Warm-rain autoconversion, accretion, evaporation	Kessler (1969)
Sundqvist large-scale condensation	Prognostic cloud water with diagnostic precipitation	Sundqvist et al. (1989, MWR); ECMWF/ECHAM lineage
Morrison double-moment	Two-moment ice–liquid with five hydrometeor categories	Morrison et al. (2005, JAS); NCAR/WRF
Seifert–Beheng	Two-moment warm + ice microphysics	Seifert & Beheng (2006, Meteorol. Atmos. Phys.)
Thompson	Mixed-phase with aerosol-aware option	Thompson et al. (2008, MWR); WRF/NOAA

Extended Data Table 4 | Boundary-layer turbulence schemes.

Scheme	Description	Community lineage
Louis	First-order stability-dependent $K(Ri)$	Louis (1979, Boundary-Layer Meteorol.)
Holtstlag–Boville	Nonlocal countergradient transport	Holtstlag & Boville (1993, J. Climate); NCAR/CESM
YSU	Nonlocal K -profile with explicit entrainment	Hong et al. (2006, MWR); WRF/Yonsei
TKE 1.5-order	Prognostic TKE with diagnostic length scale	Mellor & Yamada (1982, Rev. Geophys.)
CLUBB-lite	Higher-order closure (simplified)	Golaz et al. (2002, JAS); CESM/E3SM
Smagorinsky	Diagnostic eddy viscosity for LES/CRM	Smagorinsky (1963, MWR)

Extended Data Table 5 | Gravity-wave drag schemes.

Scheme	Description	Community lineage
Lindzen	Orographic and convective gravity-wave breaking	Lindzen (1981, JGR)
Hines	Doppler-spread non-orographic GWD	Hines (1997, JAS); ECHAM/DLR
McFarlane	Orographic wave breaking with saturation	McFarlane (1987, JAS); CESM/CCCma
Rayleigh	Sponge-layer damping	Standard implementation

Extended Data Table 6 | Ocean physics parameterizations.

Scheme	Description	Community lineage
KPP	<i>K</i> -profile parameterization for the surface boundary layer	Large et al. (1994, Rev. Geophys.); NCAR/CESM/MOM
Gent–McWilliams	Isopycnal eddy transport with skew flux	Gent & McWilliams (1990, JPO)
Redi diffusion	Along-isopycnal tracer diffusion	Redi (1982, JPO)
Visbeck adaptive GM	Flow-dependent GM coefficient	Visbeck et al. (1997, JPO)
Jayne–St Laurent tidal mixing	Abyssal tidal dissipation parameterization	Jayne & St Laurent (2001, GRL)
Split-explicit barotropic	Free-surface sub-cycling	Shchepetkin & McWilliams (2005, Ocean Model.)

Extended Data Table 7 | Land-surface and carbon-cycle schemes.

Scheme	Description	Community lineage
Slab energy balance	Single-layer energy balance with bucket hydrology	Manabe (1969, MWR)
Multi-layer Richards	Mixed-form Picard solver with six retention curves	van Genuchten (1980, SSSAJ); CLM/CESM lineage
DALEC carbon	Six-pool carbon cycle (labile/foilage/root/wood/litter/SOM)	Williams et al. (2005, Global Change Biol.)
Farquhar photosynthesis	C3 biochemical model with stomatal conductance	Farquhar et al. (1980, Planta); Medlyn et al. (2011, Global Change Biol.)

Extended Data Table 8 | Sea-ice schemes.

Scheme	Description	Community lineage
Zero-layer thermodynamics	Semtner zero-layer ice–ocean heat balance	Semtner (1976, JPO)
EVP/mEVP rheology	Elastic–viscous–plastic ice dynamics	Hunke & Dukowicz (1997, JPO); Bouillon et al. (2013); CICE lineage
Multi-category ITD	Ice-thickness distribution with linear remapping	Lipscomb (2001, JGR); CICE/CESM

Extended Data Table 9 | Coupler and surface-flux schemes.

Scheme	Description	Community lineage
COARE 3.0 bulk fluxes	State-of-the-art air–sea bulk-flux algorithm	Fairall et al. (2003, J. Climate)
Large & Yeager bulk fluxes	Alternative bulk flux for climate-model coupling	Large & Yeager (2004, NCAR Tech. Note); CESM
Briegleb ocean albedo	Zenith-angle-dependent ocean surface albedo	Briegleb et al. (1986); Briegleb (1992)

Extended Data Table 10 | Machine learning and data assimilation.

Scheme	Description	Community lineage
SFNO (spherical FNO)	Spherical Fourier neural operator for learned dynamics on the sphere	Bonev et al. (2023, ICML); Pathak et al. (2022), FourCastNet
NeuralGCM coupling	Learned atmospheric emulator coupled to legoESM physics and ocean	Kochkov et al. (2024, Nature)
4D-Var DA	Incremental variational assimilation with exact adjoint from reverse-mode AD	Courtier et al. (1994, QJRM)

References

- [1] V. Eyring, S. Bony, G. A. Meehl, C. A. Senior, B. Stevens, R. J. Stouffer, and K. E. Taylor. Overview of the Coupled Model Intercomparison Project Phase 6 (CMIP6) experimental design and organization. *Geoscientific Model Development*, 9:1937–1958, 2016. doi: 10.5194/gmd-9-1937-2016.
- [2] IPCC. *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, 2021. doi: 10.1017/9781009157896.
- [3] S. C. Sherwood, M. J. Webb, J. D. Annan, K. C. Armour, P. M. Forster, J. C. Hargreaves, G. Hegerl, S. A. Klein, K. D. Marvel, E. J. Rohling, et al. An assessment of Earth’s climate sensitivity using multiple lines of evidence. *Reviews of Geophysics*, 58(4):e2019RG000678, 2020. doi: 10.1029/2019RG000678.
- [4] P. A. O’Gorman. Precipitation extremes under climate change. *Current Climate Change Reports*, 1:49–59, 2015. doi: 10.1007/s40641-015-0009-3.
- [5] V. K. Arora, A. Katavouta, R. G. Williams, C. D. Jones, V. Brovkin, P. Friedlingstein, et al. Carbon–concentration and carbon–climate feedbacks in CMIP6 models and their comparison to CMIP5 models. *Biogeosciences*, 17:4173–4222, 2020. doi: 10.5194/bg-17-4173-2020.
- [6] P. Friedlingstein, M. O’Sullivan, M. W. Jones, R. M. Andrew, et al. Global carbon budget 2022. *Earth System Science Data*, 14:4811–4900, 2022. doi: 10.5194/essd-14-4811-2022.
- [7] P. Gentine, M. Pritchard, S. Rasp, G. Reinaudi, and G. Yacalis. Could machine learning break the convection parameterization deadlock? *Geophysical Research Letters*, 45:5742–5751, 2018. doi: 10.1029/2018GL078202.
- [8] S. Rasp, M. S. Pritchard, and P. Gentine. Deep learning to represent subgrid processes in climate models. *Proceedings of the National Academy of Sciences*, 115:9684–9689, 2018. doi: 10.1073/pnas.1810286115.
- [9] N. D. Brenowitz and C. S. Bretherton. Prognostic validation of a neural network unified physics parameterization. *Geophysical Research Letters*, 45:6289–6298, 2018. doi: 10.1029/2018GL078510.
- [10] J. Yuval and P. A. O’Gorman. Stable machine-learning parameterization of subgrid processes for climate modeling at a range of resolutions. *Nature Communications*, 11:3295, 2020. doi: 10.1038/s41467-020-17142-3.

- [11] T. Bolton and L. Zanna. Applications of deep learning to ocean data inference and subgrid parameterization. *Journal of Advances in Modeling Earth Systems*, 11:376–399, 2019. doi: 10.1029/2018MS001472.
- [12] L. Zanna and T. Bolton. Data-driven equation discovery of ocean mesoscale closures. *Geophysical Research Letters*, 47:e2020GL088376, 2020. doi: 10.1029/2020GL088376.
- [13] F. Iglesias-Suarez, P. Gentine, B. Solino-Fernandez, T. Beucler, M. Pritchard, J. Runge, and V. Eyring. Causally-informed deep learning to improve climate models and projections. *Journal of Geophysical Research: Atmospheres*, 129:e2023JD039202, 2024. doi: 10.1029/2023JD039202.
- [14] A. Grundner, T. Beucler, J. Savre, A. Lauer, M. Schlund, and V. Eyring. Reduced cloud cover errors in a hybrid AI-climate model through equation discovery and automatic tuning. *Scientific Reports*, 15:43836, 2025. doi: 10.1038/s41598-025-29155-3.
- [15] H. Heuer, T. Beucler, M. Schwabe, J. Savre, M. Schlund, and V. Eyring. Beyond the training data: confidence-guided mixing of parameterizations in a hybrid AI-climate model. *Journal of Advances in Modeling Earth Systems*, 18:e2025MS005544, 2026. doi: 10.1029/2025MS005544.
- [16] J. Ott, M. Pritchard, N. Best, E. Linstead, M. Curcic, and P. Baldi. A Fortran–Keras deep learning bridge for scientific computing. *Scientific Programming*, 2020:8888811, 2020. doi: 10.1155/2020/8888811.
- [17] F. Hourdin, T. Mauritsen, A. Gettelman, J.-C. Golaz, V. Balaji, Q. Duan, D. Folini, D. Ji, D. Klocke, Y. Qian, et al. The art and science of climate model tuning. *Bulletin of the American Meteorological Society*, 98:589–602, 2017. doi: 10.1175/BAMS-D-15-00135.1.
- [18] P. Bonnet, L. Pastori, M. Schwabe, M. Giorgetta, F. Iglesias-Suarez, and V. Eyring. Tuning the ICON-A 2.6.4 climate model with machine-learning-based emulators and history matching. *Geoscientific Model Development*, 18:3681–3706, 2025. doi: 10.5194/gmd-18-3681-2025.
- [19] E. Cleary, A. Garbuno-Inigo, S. Lan, T. Schneider, and A. M. Stuart. Calibrate, emulate, sample. *Journal of Computational Physics*, 424:109716, 2021. doi: 10.1016/j.jcp.2020.109716.
- [20] F. Couvreux, F. Hourdin, D. Williamson, R. Roehrig, V. Volodina, N. Villefranque, C. Rio, O. Audouin, J. Salter, E. Bazile, et al. Process-based climate model development harnessing machine learning: I. a calibration tool for parameterization improvement. *Journal of Advances in Modeling Earth Systems*, 13:e2020MS002217, 2021. doi: 10.1029/2020MS002217.
- [21] T. Schneider, S. Lan, A. Stuart, and J. Teixeira. Earth system modeling 2.0: A blueprint for models that learn from observations and targeted high-resolution simulations. *Geophysical Research Letters*, 44:12396–12417, 2017. doi: 10.1002/2017GL076101.
- [22] O. Fuhrer, T. Chadha, T. Hoefler, G. Kwasniewski, X. Lapillonne, D. Leutwyler, D. Lüthi, C. Osuna, C. Schär, T. C. Schulthess, and H. Vogt. Near-global climate simulation at 1 km resolution: establishing a performance baseline on 4888 GPUs with COSMO 5.0. *Geoscientific Model Development*, 11:1665–1681, 2018. doi: 10.5194/gmd-11-1665-2018.
- [23] P. Bauer, B. Stevens, and W. Hazeleger. A digital twin of Earth for the green transition. *Nature Climate Change*, 11:80–83, 2021. doi: 10.1038/s41558-021-00986-y.

- [24] V. Eyring, P. Gentine, G. Camps-Valls, D. M. Lawrence, and M. Reichstein. AI-empowered next-generation multiscale climate modelling for mitigation and adaptation. *Nature Geoscience*, 17:963–971, 2024. doi: 10.1038/s41561-024-01527-w.
- [25] I. M. Held. The gap between simulation and understanding in climate modeling. *Bulletin of the American Meteorological Society*, 86:1609–1614, 2005. doi: 10.1175/BAMS-86-11-1609.
- [26] N. Jeevanjee, P. Hassanzadeh, S. Hill, and A. Sheshadri. A perspective on climate model hierarchies. *Journal of Advances in Modeling Earth Systems*, 9:1760–1771, 2017. doi: 10.1002/2017MS001038.
- [27] P. Maher, E. P. Gerber, B. Medeiros, T. M. Merlis, S. Sherwood, et al. Model hierarchies for understanding atmospheric circulation. *Reviews of Geophysics*, 57:250–280, 2019. doi: 10.1029/2018RG000607.
- [28] J. Pathak, S. Subramanian, P. Harrington, S. Raja, A. Chattopadhyay, M. Mardani, T. Kurth, D. Hall, Z. Li, K. Azizzadenesheli, P. Hassanzadeh, K. Kashinath, and A. Anandkumar. Four-CastNet: A global data-driven high-resolution weather model using adaptive Fourier neural operators. *arXiv preprint arXiv:2202.11214*, 2022. doi: 10.48550/arXiv.2202.11214.
- [29] K. Bi, L. Xie, H. Zhang, X. Chen, X. Gu, and Q. Tian. Accurate medium-range global weather forecasting with 3D neural networks. *Nature*, 619:533–538, 2023. doi: 10.1038/s41586-023-06185-3.
- [30] R. Lam, A. Sanchez-Gonzalez, M. Willson, P. Wirsberger, M. Fortunato, F. Alet, S. Ravuri, T. Ewalds, Z. Eaton-Rosen, W. Hu, et al. Learning skillful medium-range global weather forecasting. *Science*, 382:1416–1421, 2023. doi: 10.1126/science.adi2336.
- [31] F. Lehmann, F. Ozdemir, Y. Cheng, T. Hoefler, S. Schemm, B. Soja, and S. Mishra. Can AI weather models predict beyond two weeks? A quantitative benchmark and analysis of long rollouts. *arXiv preprint arXiv:2605.30184*, 2026. doi: 10.48550/arXiv.2605.30184.
- [32] D. Kochkov, J. Yuval, I. Langmore, P. Norgaard, J. Smith, G. Mooers, M. Klöwer, J. Lottes, S. Rasp, P. Düben, et al. Neural general circulation models for weather and climate. *Nature*, 632:1060–1066, 2024. doi: 10.1038/s41586-024-07744-y.
- [33] Ellen H. Davenport, J. Varan Madan, and Duncan Watson-Parris. JCM v1.0: A differentiable, intermediate-complexity atmospheric model. *EGUsphere [preprint]*, 2026. doi: 10.5194/egusphere-2025-6266.
- [34] M. Klöwer, M. Gelbrecht, D. Hotta, J. Willmert, S. Silvestri, G. L. Wagner, A. White, S. Hatfield, T. Kimpson, N. C. Constantinou, and C. Hill. SpeedyWeather.jl: Reinventing atmospheric general circulation models towards interactivity and differentiability. *Journal of Open Source Software*, 9(98):6323, 2024. doi: 10.21105/joss.06323.
- [35] D. Häfner, R. L. Jacobsen, C. Eden, M. R. B. Kristensen, M. Jochum, R. Nuterman, and B. Vinter. Veros v0.1 – a fast and versatile ocean simulator in pure Python. *Geoscientific Model Development*, 11:3299–3312, 2018. doi: 10.5194/gmd-11-3299-2018.
- [36] J. Emmanuel Johnson. jaxsw: Differentiable approximate ocean models in JAX. <https://github.com/jejjohnson/jaxsw>, 2023.

- [37] William S. Moses et al. DJ4Earth: Differentiable and performance-portable earth system modeling via program transformations. *Journal of Advances in Modeling Earth Systems*, 2026.
- [38] Peishi Jiang et al. JAX-CanVeg: A differentiable land surface model. *Water Resources Research*, 61(1), 2025. doi: 10.1029/2024WR038116.
- [39] C. Shen, A. P. Appling, P. Gentine, T. Bandai, H. Gupta, A. Tartakovsky, M. Baity-Jesi, F. Fenicia, D. Kifer, L. Li, et al. Differentiable modelling to unify machine learning and physical models for geosciences. *Nature Reviews Earth & Environment*, 4:552–567, 2023. doi: 10.1038/s43017-023-00450-9.
- [40] M. Gelbrecht, A. White, S. Bathiany, and N. Boers. Differentiable programming for Earth system modeling. *Geoscientific Model Development*, 16:3123–3135, 2023. doi: 10.5194/gmd-16-3123-2023.
- [41] D. Kochkov, J. A. Smith, A. Alieva, Q. Wang, M. P. Brenner, and S. Hoyer. Machine learning–accelerated computational fluid dynamics. *Proceedings of the National Academy of Sciences*, 118:e2101784118, 2021. doi: 10.1073/pnas.2101784118.
- [42] D. L. Williamson, J. B. Drake, J. J. Hack, R. Jakob, and P. N. Swarztrauber. A standard test set for numerical approximations to the shallow water equations in spherical geometry. *Journal of Computational Physics*, 102:211–224, 1992. doi: 10.1016/S0021-9991(05)80016-6.
- [43] C. Jablonowski and D. L. Williamson. A baroclinic instability test case for atmospheric model dynamical cores. *Quarterly Journal of the Royal Meteorological Society*, 132:2943–2975, 2006. doi: 10.1256/qj.06.12.
- [44] J. Mouallem, L. M. Harris, and R. Benson. A duo-grid methodology for cubed-sphere shallow water models. *Journal of Computational Physics*, 485:112097, 2023. doi: 10.1016/j.jcp.2023.112097.
- [45] B. Bonev, T. Kurth, C. Hundt, J. Pathak, M. Baust, K. Kashinath, and A. Anandkumar. Spherical Fourier neural operators: Learning stable dynamics on the sphere. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *PMLR*, pages 2806–2823, 2023.
- [46] H. Hersbach, B. Bell, P. Berrisford, S. Hirahara, A. Horányi, J. Muñoz-Sabater, J. Nicolas, C. Peubey, R. Radu, D. Schepers, et al. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146:1999–2049, 2020. doi: 10.1002/qj.3803.
- [47] J. Smagorinsky. General circulation experiments with the primitive equations. I. the basic experiment. *Monthly Weather Review*, 91:99–164, 1963. doi: 10.1175/1520-0493(1963)091<0099:GCEWTP>2.3.CO;2.
- [48] R. J. Beare, M. K. Macvean, A. A. M. Holtslag, J. Cuxart, I. Esau, J.-C. Golaz, M. A. Jimenez, M. Khairoutdinov, B. Kosovic, D. Lewellen, et al. An intercomparison of large-eddy simulations of the stable boundary layer. *Boundary-Layer Meteorology*, 118:247–272, 2006. doi: 10.1007/s10546-004-2820-6.
- [49] B. Stevens, C.-H. Moeng, A. S. Ackerman, C. S. Bretherton, A. Chlond, S. de Roode, J. Edwards, J.-C. Golaz, H. Jiang, M. Khairoutdinov, et al. Evaluation of large-eddy simulations via observations of nocturnal marine stratocumulus. *Monthly Weather Review*, 133:1443–1462, 2005. doi: 10.1175/MWR2930.1.

- [50] M. C. van Zanten, B. Stevens, L. Nuijens, A. P. Siebesma, A. S. Ackerman, F. Burnet, A. Cheng, F. Couvreux, H. Jiang, M. Khairoutdinov, et al. Controls on precipitation and cloudiness in simulations of trade-wind cumulus as observed during RICO. *Journal of Advances in Modeling Earth Systems*, 3:M06001, 2011. doi: 10.1029/2011MS000056.
- [51] A. A. Wing, K. A. Reed, M. Satoh, B. Stevens, S. Bony, and T. Ohno. Radiative-convective equilibrium model intercomparison project (RCEMIP). *Geoscientific Model Development*, 11: 793–813, 2018. doi: 10.5194/gmd-11-793-2018.
- [52] S. Shima, K. Kusano, A. Kawano, T. Sugiyama, and S. Kawahara. The super-droplet method for the numerical simulation of clouds and precipitation: a particle-based and probabilistic microphysics model coupled with a non-hydrostatic model. *Quarterly Journal of the Royal Meteorological Society*, 135:1307–1320, 2009. doi: 10.1002/qj.441.
- [53] P. Bartman, O. Bulenok, K. Górski, A. Jaruga, G. Łazarski, M. Olesik, B. Piasecki, C. Singer, A. Talar, and S. Arabas. PySDM v1: particle-based cloud modelling package for warm-rain microphysics and aqueous chemistry. *Journal of Open Source Software*, 7(72):3219, 2022. doi: 10.21105/joss.03219.
- [54] A. F. Shchepetkin and J. C. McWilliams. The regional oceanic modeling system (ROMS): a split-explicit, free-surface, topography-following-coordinate oceanic model. *Ocean Modelling*, 9:347–404, 2005. doi: 10.1016/j.ocemod.2004.08.002.
- [55] W. G. Large, J. C. McWilliams, and S. C. Doney. Oceanic vertical mixing: A review and a model with a nonlocal boundary layer parameterization. *Reviews of Geophysics*, 32:363–403, 1994. doi: 10.1029/94RG01872.
- [56] P. R. Gent and J. C. McWilliams. Isopycnal mixing in ocean circulation models. *Journal of Physical Oceanography*, 20:150–155, 1990. doi: 10.1175/1520-0485(1990)020<0150:IMIOCM>2.0.CO;2.
- [57] G. Madec and the NEMO team. NEMO ocean engine, version 3.6. Technical report, Institut Pierre-Simon Laplace (IPSL), Note du Pôle de modélisation 27, 2017.
- [58] D. M. Lawrence, R. A. Fisher, C. D. Koven, K. W. Oleson, S. C. Swenson, G. Bonan, et al. The Community Land Model version 5: description of new features, benchmarking, and impact of forcing uncertainty. *Journal of Advances in Modeling Earth Systems*, 11:4245–4287, 2019. doi: 10.1029/2018MS001583.
- [59] D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, 2015. doi: 10.48550/arXiv.1412.6980.
- [60] K. Jordan, Y. Jin, V. Boza, J. You, F. Cesista, L. Newhouse, and J. Bernstein. Muon: An optimizer for hidden layers in neural networks. <https://kellerjordan.github.io/posts/muon/>, 2024.
- [61] I. M. Held and M. J. Suarez. A proposal for the intercomparison of the dynamical cores of atmospheric general circulation models. *Bulletin of the American Meteorological Society*, 75: 1825–1830, 1994. doi: 10.1175/1520-0477(1994)075<1825:APFTIO>2.0.CO;2.